

# NovaPharma Iberia — Step 1 Review: Data Model Discovery

Digital transformation project · Excel → MySQL / Python / Power BI / Anaplan · Author: Nix · 17 September 2026

## 1. Isolating the data

The legacy workbook *NovaPharma\_Commercial\_Model.xlsx* mixes data, lookups, calculations, reports and macro support in 17 sheets. The first step was to separate the sheets that actually hold data from everything else. Using the tab colours as a guide (grey = master data, red = sales, green = plan and market data), eight sheets were copied into a data-only workbook, *NovaPharma\_Data\_Model\_Study.xlsx*, with all formulas and derived lookup columns removed. Report sheets (Consolidation, BvA, KPI\_Dashboard, Data\_Quality), parameters and macro sheets were excluded: they are outputs or plumbing, not tables. Sales\_Enriched was also excluded, since it is Sales\_Raw plus lookups — in a database those lookups become JOINS.

## 2. Classification method

Every sheet was described with five characteristics: **type** (dimension or fact), **grain** (what one row means), **primary key**, **foreign keys** (which table each key points to) and **measures** (the numbers that get aggregated in reports). The type was decided with one rule: if the sheet answers a quantitative question (*how much did we sell? how much do we plan to sell?*) it is a fact; if it answers a qualitative one (*what, who, where, when, how?*) it is a dimension. Two kinds of fact were found: **transactional** (Sales\_Raw — one row per invoice line, with its own ID as primary key) and **snapshot** (Budget, Forecast, Market — exactly one row per combination of dimensions, so the primary key is the combination of foreign keys).

## 3. Bad practice detected and how it will be fixed

**Brand.** Market data (IQVIA) arrives at brand level, not SKU level, so it cannot point to *Products*, whose key is SKU\_Code. In Excel the link only works because the brand name is typed identically in both sheets — an implicit hierarchy held together by matching text. Fix: create an explicit dimension **dim\_brand** (brand code, brand name, therapeutic area) and make both *Products* and *Market* point to it. This is a snowflake, the same pattern already used by Customers → Territories.

**Version.** Budget and Forecast carry a *Version* column ("Budget 2026", "FC 2026-Q2") with no table behind it, and the forecast cut-off month lives only in the version name. Fix: create **dim\_version** (version name, type, cut-off month). Budget and Forecast share grain and measures, so they will be loaded into one table, **fact\_plan**.

## 4. Final inventory (source sheets + new dimensions for the target model)

| #  | Table       | Origin | Type                 | Grain (1 row = ...)                            | PK                                              | FK → table                                                                                     | Measures                             |
|----|-------------|--------|----------------------|------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------|
| 1  | Calendar    | sheet  | Dim                  | 1 month                                        | Month_Key                                       | —                                                                                              | —                                    |
| 2  | Products    | sheet  | Dim                  | 1 SKU                                          | SKU_Code                                        | Brand → dim_brand                                                                              | —                                    |
| 3  | Customers   | sheet  | Dim                  | 1 customer                                     | Customer_Code                                   | Territory_Code → Territories                                                                   | —                                    |
| 4  | Territories | sheet  | Dim                  | 1 territory                                    | Territory_Code                                  | —                                                                                              | —                                    |
| 5  | dim_brand   | new    | Dim                  | 1 brand                                        | Brand_Code                                      | —                                                                                              | —                                    |
| 6  | dim_version | new    | Dim                  | 1 plan version                                 | Version_Name                                    | —                                                                                              | —                                    |
| 7  | Sales_Raw   | sheet  | Fact (transactional) | 1 invoice line                                 | Transaction_ID                                  | SKU_Code → Products; Customer_Code → Customers; Month_Key (derived) → Calendar                 | Units, Gross_Sales, Discount, Rebate |
| 8  | Budget      | sheet  | Fact (snapshot)      | 1 plan line: SKU × territory × month × version | Version + Month_Key + SKU_Code + Territory_Code | Month_Key → Calendar; SKU_Code → Products; Territory_Code → Territories; Version → dim_version | Units, Net_Sales                     |
| 9  | Forecast    | sheet  | Fact (snapshot)      | 1 estimate line: same grain as Budget          | same 4 keys                                     | same as Budget                                                                                 | Units, Net_Sales                     |
| 10 | Market      | sheet  | Fact (snapshot)      | 1 market line: brand × territory × month       | Month_Key + Brand + Territory_Code              | Month_Key → Calendar; Territory_Code → Territories; Brand → dim_brand                          | Market_Units, Market_Value           |

Result: six dimensions and four facts (three tables in SQL, since Budget and Forecast merge into fact\_plan). Highlighted rows are dimensions that did not exist in Excel and were discovered by modelling. Next step: draw the target constellation schema and write the MySQL DDL.